PHGN 515: Computational Project 2

White Light Interferometry

Project Summary

Scanning white light interferometry (WLI) is a versatile measurement tool. WLI can be used to measure the complex refractive index of materials and forms the basis for optical coherence tomography. The goal of this project is to simulate scanning white light interferometry. Specifically, the goal is to computationally reconstruct the electric field in the frequency domain, $\tilde{E}(\omega)$, for two input time domain signal, $E_1(t)$ and $E_2(t)$.

In this project, you will reconstruct the frequency domain electric field for time domain electric fields **provided by me** in the file, "electricFields.csv". The format of the file is (as column vectors): time axis [fs], $E_1(t)$ [a.u.], $E_2(t)$ [a.u.]. You will then fit an nth order polynomial to the spectral phase you recover and use the second order coefficient to estimate material composition. Finally, you will calculate the second moment, skewness, and kurtosis of the resulting spectrum.

Scanning White Light Interferometry (WLI) Theory

Simple case: No phase or amplitude difference

We first define a Fourier transform pair as

$$\tilde{F}(\omega) = \int_{-\infty}^{\infty} f(\tau)\, e^{-i\,\omega\,\tau}\, d\tau \equiv \mathcal{F}\{f(\tau)\}$$

$$f(\tau) = \frac{1}{2\,\pi} \int_{-\infty}^{\infty} \tilde{F}(\omega)\, e^{i\,\omega\,\tau}\, d\omega \equiv \mathcal{F}^{-1}\{\tilde{F}(\omega)\}$$

Then, generally speaking, the total field in WLI is the superposition of two fields, one from either arm of the interferometer

$$E_T(t) = E_1(t) + E_2(t)$$

in the simplest case, E_2 is just a shifted version of E_1, that is to say

$$E_2(t) = E_1(t - \tau) = E(t - \tau)$$

The action of the detector is to absolute value square the total field and then integrate over some

period so that the time domain signal is

$$\boldsymbol{I(\tau) = \int_{-T/2}^{T/2} |E_T(t)|^2 \, dt \approx \int_{-\infty}^{\infty} |E(t) + E(t-\tau)|^2 \, dt}$$

THIS IS WHAT YOU SHOULD DO IN YOUR SIMULATION

Here, we have made the approximation that the integration period is extremely long e.g., $T\,\omega_0 >>> 1$, where T is the integration period and ω_0 is the mean frequency of the white light. Continuing on, expanding the absolute value squared

$$I(\tau) = \int_{-\infty}^{\infty} |E(t)|^2 + |E(t-\tau)|^2 + E(t)\,E(t-\tau)^* + E(t)^*\,E(t-\tau)\,dt$$

the first two integrals are constant, which we will define as C

$$I(\tau) = C + \int_{-\infty}^{\infty} E(t)\,E(t-\tau)^* + E(t)^*\,E(t-\tau)\,dt$$

we now write the electric field as

$$E(t) = A(t)\,e^{i\,\omega_0\,t}$$

where $A(t)$ is a complex amplitude (e.g., it can carry a phase). Substituting this into our expression,

$$I(\tau) = C + \int_{-\infty}^{\infty} A(t)\,e^{i\,\omega_0\,t}\,A^*(t-\tau)\,e^{-i\,\omega_0(t-\tau)} + A^*(t)\,e^{-i\,\omega_0\,t}\,A(t-\tau)\,e^{i\,\omega_0(t-\tau)}\,dt$$

simplifying

$$I(\tau) = C + \int_{-\infty}^{\infty} A(t)\,A^*(t-\tau)\,e^{i\,\omega_0\,\tau} + A^*(t)\,A(t-\tau)\,e^{-i\,\omega_0\,\tau}\,dt$$

it is instructive to see that when A is purely real, the signal is composed of a constant, C, and an oscillating function times the autocorrelation of the temporal envelop

$$\boldsymbol{I(\tau) = C + 2\cos(\omega_0\,\tau)\int_{-\infty}^{\infty} A(t)\,A(t-\tau)\,dt}$$

YOU SHOULD SEE SOMETHING LIKE THIS IN YOUR SIMULATION

and now taking the Fourier transform from $\tau \longleftrightarrow \omega$ by multiplying the entire expression by $e^{-i\,\omega\,\tau}$ and integrating over τ

$$\tilde{I}(\omega) = C\,\delta(\omega) + \int_{-\infty}^{\infty} A(t)\Big[\int_{-\infty}^{\infty} A^*(t-\tau)\,e^{i\,\omega_0\,\tau}\,e^{-i\,\omega\,\tau}\,d\tau\Big] + A^*(t)\Big[\int_{-\infty}^{\infty} A(t-\tau)\,e^{-i\,\omega_0\,\tau}\,e^{-i\,\omega\,\tau}\,d\tau\Big]\,dt$$

where I have defined $\tilde{I}(\omega) \equiv \mathcal{F}\{I(\tau)\}$, simplifying

$$\tilde{I}(\omega) = C\,\delta(\omega) + \int_{-\infty}^{\infty} A(t)\left[\int_{-\infty}^{\infty} A^*(t-\tau)\, e^{-i\,\tau\,(\omega-\omega_0)}\, d\tau\right] + A^*(t)\left[\int_{-\infty}^{\infty} A(t-\tau)\, e^{-i\,\tau(\omega+\omega_0)}\, d\tau\right] dt$$

we see that this is actually computing the Fourier transforms from $\tau \longleftrightarrow \omega \pm \omega_0$. Defining

$u \equiv \tau - t$ and $du = d\tau$

$$\tilde{I}(\omega) = C\,\delta(\omega) + \int_{-\infty}^{\infty} A(t)\left[e^{-i\,t\,(\omega-\omega_0)} \int_{-\infty}^{\infty} A^*(-u)\, e^{-i\,u\,(\omega-\omega_0)}\, du\right] + A^*(t)\left[e^{-i\,t\,(\omega+\omega_0)} \int_{-\infty}^{\infty} A(-u)\, e^{-i\,u\,(\omega+\omega_0)}\, du\right] dt$$

the Fourier transforms evaluate to

$$\tilde{I}(\omega) = C\,\delta(\omega) + \tilde{A}^*(\omega-\omega_0) \int_{-\infty}^{\infty} A(t)\, e^{-i\,t(\omega-\omega_0)}\, dt + \tilde{A}(-\omega-\omega_0) \int_{-\infty}^{\infty} A^*(t)\, e^{-i\,t(\omega+\omega_0)}\, dt$$

and we note the remaining integrals are Fourier transforms as well, which we evaluate to give

$$\tilde{I}(\omega) = C\,\delta(\omega) + \tilde{A}^*(\omega-\omega_0)\,\tilde{A}(\omega-\omega_0) + \tilde{A}(-\omega-\omega_0)\,\tilde{A}^*(-\omega-\omega_0)$$

which is

$$\boldsymbol{\tilde{I}(\omega) = C\,\delta(\omega) + \left|\tilde{A}(\omega-\omega_0)\right|^2 + \left|\tilde{A}(-\omega-\omega_0)\right|^2}$$

THIS IS WHAT YOU SHOULD GET IN YOUR SIMULATION

In this simple case, the Fourier transform of the time domain correlation produces two copies of the **spectrum**. One of the copies is centered at ω_0 and is not flipped. The other copy is centered at $-\omega_0$ and is flipped. This is what you will recover in your simulation.

General case: Amplitude & Phase difference at difference central frequencies

The full derivation for this case is relatively involved but follows the same template as the simple case. The difference in the current derivation is to write each field as a complex amplitude times the carrier with different complex amplitudes and different central frequencies:

$$E_j(t) = A_j(t)\, e^{i\,\omega_j\,t} \qquad j = 1, 2$$

The result of Fourier transforming the cross-correlation between two electric fields is

$$\tilde{I}(\omega) = C\,\delta(\omega) + \tilde{A}_1(\omega-\omega_1)\,\tilde{A}_2(\omega-\omega_2)^* + \tilde{A}_1(-\omega-\omega_1)^*\,\tilde{A}_2(-\omega-\omega_2)$$

This is the general result that admits different spectral amplitudes and center frequencies. Now, for instance, if $\omega_1 = \omega_2 \equiv \omega_0$

$$\tilde{I}(\omega) = C\,\delta(\omega) + \tilde{A}_1(\omega - \omega_0)\,\tilde{A}_2(\omega - \omega_0)^* + \tilde{A}_1(-\omega - \omega_0)^*\,\tilde{A}_2(-\omega - \omega_0)$$

and even more specifically if $\tilde{A}_1 = \tilde{A}_2 \equiv \tilde{A}$

$$\tilde{I}(\omega) = C\,\delta(\omega) + \left|\tilde{A}(\omega - \omega_0)\right|^2 + \left|\tilde{A}(-\omega - \omega_0)\right|^2$$

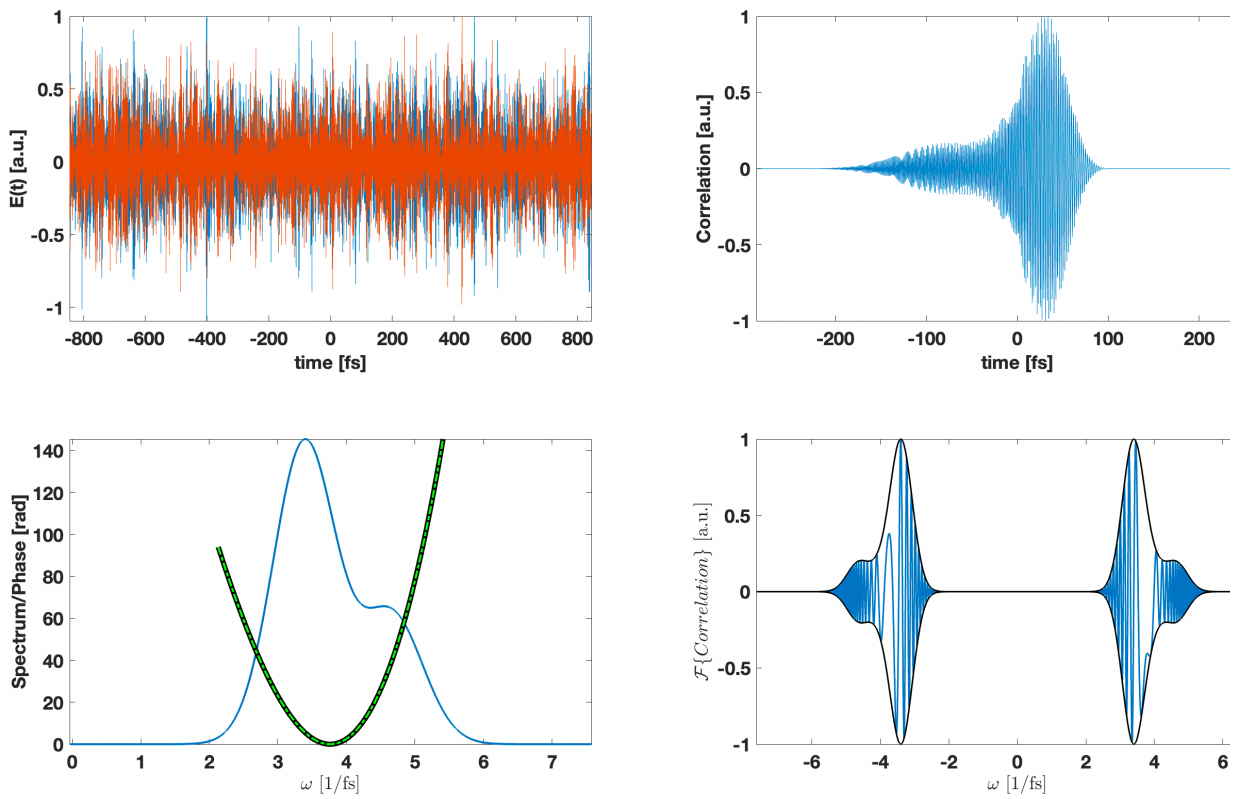
In this last, illustrative case, we see a spike at DC and then two AC terms with one reversed along the ω-axis. Both terms appear at ω_0, one at a positive frequency and one at a negative frequency. In the general case, the appearance of terms that go like $\tilde{A}_2(\omega - \omega_2)^*\,\tilde{A}_1(\omega - \omega_1)$ means that the frequency bands of the two signals have to overlap or they simply won't interfere.

Additionally, the product of the two fields means that the ***phase difference*** is obtained in addition to the **product of the spectra**. The fact that the phase difference is recovered is vital to the operation of a white light interferometer.

Mirrors and other optics can impart spectral phase to the fields so a calibration must be done to remove phase differences caused by the device itself. After calibrating phase differences from the device, a nominally transparent material can be placed in one arm of the interferometer. Then any phase difference between the two fields in the interferometer is only the phase imparted by the material. This means that the dispersion of transparent materials can be recovered using scanning white light interferometry.

Deliverable

Using the electric fields provided with this assignment generate the following plot, comprised of 4 plots



The plots show (counting clockwise from top left):

1. Real part of both time domain electric fields (on same plot) [red, blue]
2. Circular cross-correlation of these fields
3. Fourier transform of the correlation
4. Spectrum [blue solid] and phase [black and dashed green] of spectral domain electric field

Note that the reconstruction process i.e., the Fourier transform from correlation to frequency space, produces two copies of the frequency domain product field [Plot 3] above e.g.,

$$\tilde{A}_1(\omega-\omega_0)\,\tilde{A}_2(\omega-\omega_0)^* + \tilde{A}_1(-\omega-\omega_0)^*\,\tilde{A}_2(-\omega-\omega_0)$$

You should show only one spectral amplitude and one spectral phase [Plot 4]. In this assignment, you will assume that the electric fields provided differ only by phase, and have the same amplitudes.

Polynomial fit to the phase:

You will fit an nth order polynomial to the spectral phase you recover. The spectral phase may be represented by its Taylor series expansion about $\omega = \omega_0$:

$$\varphi(\omega-\omega_0) \approx \varphi(0) + \partial_\omega \varphi(0)\,(\omega-\omega_0) + \frac{1}{2}\,\partial_\omega^2 \varphi(0)\,(\omega-\omega_0)^2 + \ldots + \frac{1}{n!}\,\partial_\omega^n \varphi(0)\,(\omega-\omega_0)^n$$

In this case, the second order coefficient $\partial_\omega^2 \varphi(0)$ is known as the Group Delay Dispersion (GDD) in units

of fs^2. Assume the phase difference is due to the dispersion of a 1 mm thick window placed in one arm of the white light interferometer. Match the value of your second order coefficient (from your polynomial fit) to one of the materials in the 'popular optical glasses' section on refractiveindex.info. **Report what material the window is made from**. **Also report the R^2 value from your fit.**

Note: Group Velocity Dispersion (GVD) is the GDD per millimeter e.g., GVD [=] GDD/mm

Answer the following questions:

1. Comment on the structure of the reconstructed, frequency-dependent field given the two time dependent fields supplied to you with this exercise. Specifically, provide an explanation for the seemingly random appearance of the time dependent fields, given your reconstruction.

2. Provide an explanation for the asymmetry of the cross-correlation function.

3. Propose an explanation for the relative strengths of the polynomial coefficients from your fit. Specifically, comment on the fact that the dominant term is parabolic given the experimental configuration.

Grading Rubric

PHGN515 Rubric	Far Exceeds Expectations	Exceeds Expectations	Meets Expectations	Partially Meets Expectations	Does Not Meet Expectations
Grade Value	[5]	[4]	[3]	[2]	[1]
Support & Evidence	**Four plots showing:** 1. Real part of both time domain electric fields 2. Cross–correlation from these fields 3. Fourier transform of correlation 4. Spectrum and phase **Superior execution entials:** (1) Functions labeled with colors and styles (2) Appropriate labels: axes, title, and legend (3) Visually, clean and organized **Correct:** 1. window material 2. second moment, skewness, kurtosis 3. explanations to questions	**Four plots showing:** 1. Real part of both time domain electric fields 2. Cross–correlation from these fields 3. Fourier transform of correlation 4. Spectrum and phase **Execution entials:** (2) Appropriate labels: axes, title, and legend (3) Visually, clean and organized **Correct:** 1. window material 2. second moment, skewness, kurtosis 3. explanations to questions	**Four plots showing:** 1. Real part of both time domain electric fields 2. Cross–correlation from these fields 3. Fourier transform of correlation 4. Spectrum and phase **Execution entials:** (2) Appropriate labels: axes, title, and legend (3) Visually, clean and organized **Correct:** 1. window material 2. second moment, skewness, kurtosis	**Four plots showing:** 1. Real part of both time domain electric fields 2. Cross–correlation from these fields 3. Fourier transform of correlation 4. Spectrum and phase **Execution entials:** (3) Visually, clean and organized **Correct:** 2. second moment, skewness, kurtosis	Student handed in a smooth stone they found while on a walk
Organization	Publication quality with well thought out ideas grouped into logical sections. Code is organized with descriptive variable names, well commented, and includes a header	Graduate level quality with well thought out ideas group into logical sections. Code is organized, well commented, and includes a header	Ideas are organized into blocks but not organized in a contiguous manner. Code is organized, few comments, and does not include a header	Ideas are scattered. Code not organized—spaghetti code. no comments, no header, poor variable names	Student did not turn in working code. Student did not attempt to get help
Communication	Uses expert level language and domain–specific vocabulary drawing on ideas from resources including class material and text. Publication quality grammar, punctuation, appearence, and citations. Student motivates arguments with concise calculations that highlight key concepts.	Graduate level textual descriptions including sources and some mathematical derivation. Student uses some domain–secific vocabulary and draws on ideas from resources including class material and text. Appropriated grammar, punctuation, and appearance	Good textual descriptions with a few sources. Limited motivation for conclusions with some grammar or punctuation mistakes.	Few descriptions and limited sources. No motivation for conclusion with several grammar and punctuation mistakes	Poor writing, littered with grammatical mistakes. Student did not convey any working knowledge of basic principles. Student did not attempt to get help.

Code Outline (pseudo-code)

1. Calculate circular cross-correlation of provided electric fields
2. Calculate frequency grid from provided temporal grid using the relation $d\omega = \frac{2\pi}{N\,dt}$, where N is the

number of points

3. Fourier transform correlation signal to produce frequency domain field

4. Extract the phase and amplitude of one of the two frequency domain signals

5. Fit an nth order polynomial to the phase where the amplitude is above some threshold

[Hint: The phase of a function is not very meaningful when the function is zero or close to zero]

6. Calculate the GDD from the second order coefficient and match with a material on refractiveindex.info

7. Calculate the second moment, skewness, and kurtosis of spectrum